

Curriculum Vitae – Akash Anand

PhD Student & NASA FINESST Fellow

Department of Forestry and Wildlife Ecology, University of Wisconsin-Madison

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Education

- PhD in Forestry; Department of Forestry and Wildlife Ecology 2021 - present
University of Wisconsin-Madison, USA
Advisor: Volker C. Radeloff | SILVIS Lab
- Integrated Bachelor and Master in Geoinformatics, Department of 2014 – 2019
Geoinformatics, Central University of Jharkhand, India

Research experience

- Junior Research Fellow, Institute of Environment and Sustainable 2019 - 2021
Development, Banaras Hindu University, India
Project: Hyperspectral Imaging for sharper definitions of the Himalayan ecosystems and its high value plant species under climate uncertainties

Awards and Fellowships

- NASA Future Investigator in NASA Earth and Space Science and 2024-2027
Technology (FINESST) Fellowship (\$150,000).
- Helen H. Miller Award by Graduate Committee at UW-Madison (\$1000) 2026
- NASA-MSU Professional Enhancement Award (\$800). 2025
- Thomas O'Brien Award by Graduate Committee at UW-Madison 2025
(\$1000).
- UW-Madison Student Research Grant competition (\$600). 2025
- IEEE GRSS IADF School on Computer Vision for Earth Observation 2023
Travel Award (500€).

Publications

1. **Anand, A.**, B. Deneu, V. C. Radeloff (2026). Convolutional neural networks outperform other presence-only species distribution modeling algorithms. *Proceedings of the National Academy of Sciences (PNAS)*.
2. Liu D., E. M. Silveira, E. Razenkova, **A. Anand**, ..., V. C. Radeloff (2025). Global dynamic habitat indices (DHIs) based on MODIS and VIIRS vegetation products. *Remote Sensing of Environment*.
3. Silveira, E. M. O., **A. Anand**, A. M. Pidgeon, E. Wood, R. E. Buron, B. Bar-Massada, L. S. Farwell, A. R. Ives, B. Zuckerberg, and V. C. Radeloff (2025). Scalogram habitat measures to predict bird abundance. *Ecography*.

4. Schug, F. N. K. Kasraee, **A. Anand**, A. Rizayeva, M. T. Gorth-Price, M. D. Nita, and V. C. Radeloff (2025). Quantifying urban growth with 1970s Hexagon spy satellite imagery and Mask R-CNN building footprint detection. *Landscape and Urban Planning*.
5. Razenkova, E., K. E. Lewinska, **A. Anand**, H. Yin, L. S. Farwell, A. M. Pidgeon, P. Hostert, N. C. Coops, V. C. Radeloff (2025). Medium-resolution Dynamic Habitat Indices from Landsat and Sentinel-2 satellite imagery. *Ecological Indicators*.
6. Olah, A., V. C. Radeloff, **A. Anand**, E. M. O. Silviera, N. Politi, L. Rivera, S. Martinuzzi, G. M. Pastur, A. M. Pidgeon (2025). Landscape scale effects of primary productivity on forest bird species occurrence and abundance in Argentina. *Landscape Ecology*.
7. Sagar HSSC, **Anand, A.**, Persche, M., Pidgeon, A.M., Zuckerberg, B., Sekerciaglu, C.H., Burivilova, Z. (2024). Global analysis of acoustic frequency characteristics in birds. *Proceddings of the Royal Society B*.
8. **Anand, A.**, Srivastava, P.K., Pandey, P.C., Khan, M.L., & Behera, M.D. (2022). Assessing the niche of Rhododendron arboreum using entropy and machine learning algorithms: role of atmospheric, ecological, and hydrological variables. *Journal of Applied Remote Sensing*, 16, 042402.
9. Şandric, I., Irimia, R., Petropoulos, G.P., **Anand, A.**, Srivastava, P.K., Pleşoianu, A., Faraslis, I., Stateras, D., & Kalivas, D. (2022). Tree's detection & health's assessment from ultra-high resolution UAV imagery and deep learning. *Geocarto International*, 1-21.
10. **Anand, A.**, Malhi, R.K.M., Srivastava, P.K., Singh, P., Mudaliar, A.N., Petropoulos, G.P., & Kiran, G.S. (2022). Optimal band characterization in reformation of hyperspectral indices for species diversity estimation. *Physics and Chemistry of the Earth, Parts A/B/C*, 126, 103040.
11. Malhi, R.K.M., **Anand, A.**, Srivastava, P.K., Chaudhary, S.K., Pandey, M.K., Behera, M.D., Kumar, A., Singh, P., & Kiran, G.S. (2022). Synergistic evaluation of Sentinel 1 and 2 for biomass estimation in a tropical forest of India. *Advances in Space Research*, 69, 1752-1767.
12. Malhi, R.K.M., Pandey, M.K., **Anand, A.**, Srivastava, P.K., Petropoulos, G.P., Singh, P., Sandhya Kiran, G., & Bhattacharya, B. (2022). Band selection algorithms for foliar trait retrieval using AVIRIS-NG: a comparison of feature based attribute evaluators. *Geocarto International*, 37, 4071-4087.
13. **Anand, A.**, A. S. Dinesh, P. K. Srivastava, S. K.r Choudhary, A. K. Verma, P. Kumar (2021). Rainfall Rate estimation over India using Global Precipitation Measurement's Microwave Imager dataset and Machine Learning algorithms. *Geocarto International*, Taylor & Francis.
14. Maurya. S., P. K. Srivastava, A. Yaduvanshi, **A. Anand**, G. P. Petropoulous, L. Zhuo, R. K. Mall (2022). Rainfall rate estimation over India using global precipitation measurement's microwave imager datasets and different variants of fuzzy information system. *Geocarto International*, 37(21), 6213-6231.
15. **Anand, A.**, Pandey, P.C., Petropoulos, G.P., Pavlides, A., Srivastava, P.K., Sharma, J.K., & Malhi, R.K.M. (2020). Use of hyperion for mangrove forest carbon stock assessment in Bhitarkanika forest reserve: A contribution towards blue carbon initiative. *Remote Sensing*, 12, 597.

16. Malhi, R.K.M., **Anand, A.**, Mudaliar, A.N., Pandey, P.C., Srivastava, P.K., & Sandhya Kiran, G. (2020). Synergetic use of in situ and hyperspectral data for mapping species diversity and above ground biomass in Shoolpaneshwar Wildlife Sanctuary, Gujarat. *Tropical Ecology*, 61, 106-115.

Book Chapters

- Anand, A.**, P Singh, P. K. Srivastava, M. Gupta (2021). GIS-based analysis for soil moisture estimation via kriging with external drift. In *Agricultural water management* (pp. 391-408). Academic Press.
- Anand, A.** (2022). Sentinel SAR Data and In-Situ-Based High-Resolution Above-Ground Carbon Stocks Estimation Within the Open Forests of Ramgarh District. In *Research Anthology on Ecosystem Conservation and Preserving Biodiversity* (pp. 402-422). IGI Global.
- P. K. Srivastava, R. K. M. Malhi, P. C. Pandey, **A. Anand**, P. Singh, M. K. Pandey, A. Gupta (2020). Revisiting hyperspectral remote sensing: Origin, processing, applications and way forward. In *Hyperspectral remote sensing* (pp. 3-21). Elsevier.
- Singh, A., **A. Anand**, P. K. Srivastava, A. Singh, P. C. Pandey. (2021). Delineation of groundwater potential zone and site suitability of rainwater harvesting structures using remote sensing and in situ geophysical measurements. *Advances in Remote Sensing for Natural Resource Monitoring*, 170-188, Elsevier.

Seminar/conferences

- Anand A.**, B. Deneu, V. C. Radeloff. Convolutional Neural Networks outperform all other Species Distribution Models. NASA Biodiversity and Ecological Conservation Science Team Meeting, May 27-29, 2025, Washington DC.
- Anand A.**, D. Liu, V. C. Radeloff, E. Silveira, E. Razenkova, A.M. Pidgeon, N. Coops, M. Hobi, A. Ives. The Dynamic Habitat Indices from Terra, Aqua, Suomi NPP and JPSS data for biodiversity science and conservation. NASA Biodiversity and Ecological Conservation Science Team Meeting, May 27-29, 2025, Washington DC.
- Anand A.**, M. Persche, A. Pidgeon, V. C. Radeloff. Explainable AI for Multi-Scale Insights into Bird Habitat Selection. International Association for Landscape Ecology – North America (IALE-NA), April 13-17, 2025, Raleigh, North Carolina.
- Wirth, P., A. Syphard, G. P. MacKenzy, **A. Anand**, E. Silveira, M. Mockrin, V. C. Radeloff. Factors and Spatial Scales Associated with Building Loss in Wetland Fires. Center for Ecology and the Environment Spring Symposium, April 14-15, 2025, Madison, Wisconsin.
- Radeloff, V. C., E. Silveira, R. Buron, **A. Anand**, A. M. Pidgeon, B. Zuckerberg, A. Ives, L. Farwell, A. Bar-Massada, N. Coops, and M. Hobi. Multiple spatial scales, long-term trends, and synchrony of the dynamic habitat indices and bird populations. NASA Biodiversity and Ecological Forecasting Science Team Meeting, May 6-9, 2024, Silver Spring, Maryland.
- Nasanbat, E., K. E. Lewinska, **A. Anand**, M. Nyamtseren, O. Lkhamjav, A. Dorjusen, and V. C. Radeloff. Grassland degradation in Mongolia based on 37 years of Landsat data, spectral unmixing and pathway analysis of cumulative endmember fractions. American Geophysical Union Fall Meeting. December 9-13, 2024, Washington DC.
- Liu, D., F. Schug, **A. Anand**, C. Stephens, A. Rizayeva, and V. C. Radeloff. 2024. Trends of

- global human activity and their impact on vegetation greenness. American Geophysical Union Fall Meeting. December 9-13, 2024, Washington DC.
- Olah, A.M., V. C. Radeloff, **A. Anand**, E. M. O. Silveira, N. Politi, L. Rivera, S. Martinuzzi, G. M. Pastur, A. M. Pidgeon, "Landscape scale effects of dynamic habitat indices on bird species richness and abundance in Argentina", Annual meeting of Ecological Society of America 2024, Long Beach, California, USA.
- Anand, A.**, V. C. Radeloff, A. Pidgeon, R. Buron, E. Razenkova, J. Timmer, "Implementation of Explainable AI to Map Hierarchical Habitat Selection in Birds, IEEE International Geosciences and Remote Sensing Symposium, 2023, Pasadena, California, USA.
- Radeloff, V. C., E. Silveira, R. Buron, **A. Anand**, A. M. Pidgeon, B. Zuckerberg, A. Ives, L. Farwell, A. Bar-Massada, N. Coops, and M. Hobi. 2023. Multiple spatial scales, long-term trends, and synchrony of the dynamic habitat indices and bird populations. NASA Biodiversity and Ecological Forecasting Science Team Meeting, May 8-11, 2023, College Park, Maryland.
- Anand, A.**, S. K. Chaudhary, M. K. Pandey, A. Gupta, P. Singh, P. K. Srivastava, "Estimation and Generation of Carbon Stock Product using Machine Learning Algorithm through Invocation of Cloud-Based Web Services", at ISRS-ISG National Symposium, 2020, SAC-ISRO, Ahmedabad India.
- Anand, A.**, P. K. Srivastava, P. C. Pandey, and R. K. M. Malhi, "Earth Observation and Analytical Hierarchy Process based fusion technique to model Regional Forest Fire Risk", Vol. 21, EGU2019-12423, EGU General Assembly 2019, Vienna.
- Anand, A.**, P. K. Srivastava, K. Lal, P. C. Pandey (2018), "Estimation of Above Ground Biomass for assessing the total carbon stock within the forest cover of Ramgarh District using Sentinel 1 SAR data", International Conference on Environmental Challenges and Sustainability (ICECS), 2018, Ranchi, India.

Invited Lecture/Training

- Geo-Webinar on "Introduction to Convolutional Neural Networks (CNNs) for Satellite data and Ecology", organized by SILVIS Lab at UW-Madison and Mongolian Geospatial Association, 18th March, 2025.
- Delivered a hand-on-training at the "Online Workshop on Species Distribution Modelling using R", held at the Institute of Environment and Sustainable Development, Banaras Hindu University, 15 – 16 March 2021.

Internships / trainings / workshop

- Participated in a 10-day training on "Applied Deep Learning, Computer Vision & NLP with Python" organized by Eduxlabs, E-Cell IIT Hyderabad, 16 – 27 February 2021.
- Participated in the workshop on High Performance Computing (HPC) organized by BHU, IIT (BHU) and C-DAC at DST-Centre for Interdisciplinary Mathematical Sciences (CIMS), Institute of Science, BHU, Varanasi, 20-25 January 2020.
- Participated in the 2nd National Workshop on "Techniques in Hyperspectral Data Analysis and Processing" held at Environment and Sustainable Development, jointly organized by Institute of Environment and Sustainable Development and IIT BHU, 27 - 31 January 2020.

- Attended one week training programme on “Calibration and Validation of Satellite sensors and derived products” under TREES program at Space Application Centre (SAC), ISRO Ahmedabad, India, 22 – 26 October 2018.
- Participated in National Workshop “Popularization of Remote Sensing Based Maps & Geospatial Information” jointly organized by ISRS & ISRO, August 11, 2017.